Factorisation Classes in G(6) and the Harmonics of the Primes:

Entry rule $k = \mathrm{ord}_p(3)$, chords as partitions, and the boundary of the Galois layer

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Abstract

Doc. 341 shows that GF(27) enters G(6) as an order structure through the characteristic polynomial $\chi_X = (x^3+2x+1)^2(x+1)^2$. This document asks which further factorisations χ admits in G$(6) \cong M_8$(GF(9)), and shows that the same factorisation rule organises the harmonic patterns of the primes (Doc. 057, Doc. 159).

Four results: **(A)** Over GF(3) there are eight irreducible cubic polynomials, four of them primitive (order 26) and four of order 13; they generate five factorisation classes of χ in G(6) **[B]**. **(B)** A prime p enters |GF$(3^k)^*$| exactly at $k = \mathrm{ord}_p(3)$; the harmonic primes enter in the order $13\,(k{=}3)$, $5\,(k{=}4)$, $11\,(k{=}5)$, $7\,(k{=}6)$, after which only large primes appear **[B]**. **(C)** Element orders in G(6) are partitions of 8 with an lcm rule — the algebraic form of a chord of overtones **[B]**. **(D)** From $k \geq 4$ on, the Galois grid is finer than the fractal-recursive correction ($1.05\,\%$, Doc. 338); Galois products then become statistically indistinguishable from higher torus modes. This is not a contradiction but the algebraic form of the mode convergence of Doc. 159 **[B]**.

Negative result: the tau sector is not a Galois product at $0.05\,\%$ **[S]**. Check scripts: `pruef_342_faktorisierung_primzahlen.py`, `pruef_342b_hoehere_harmonien.py`, `pruef_342c_dichte_hoehere_harmonien.py`.

.1 Starting point: algebraic equals geometric

The following status markers are used throughout:

- **[K]** *Confirmed* — algebraically or numerically fully proved; all check-script assertions passed; derived directly from ξ or a corpus-established derivation.
- **[B]** *Established* — numerically secured by sampling or explicit construction; analytical proof present or trivial.
- **[S]** *Speculative* — indication or conjecture, not yet fully proved.

Docs. 338 and 341 showed that mass ratios from torus winding numbers (r_i, p_i) and from Galois group orders yield the same number: $(m_\mu/m_e)^2 = 43200 = |\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)|$. The question whether the geometric formulation (vectors on T^4) is "more precise" than the algebraic one is ill-posed: winding numbers on S^1 and elements of GF(3^k) are both integer, exact objects. The bridge is the Frobenius automorphism $g \mapsto g^3$ on GF(3^k), which represents the $\mathbb{Z}_3$ rotation of the orbifold $T^4/\mathbb{Z}_3$ (Doc. 314, Doc. 336).

The algebraic route has one methodological advantage: polynomial factorisation can be enumerated systematically, geometric search cannot. This document exploits that.

.2 Theorem A: Factorisation classes in G(6)

Irreducible cubic polynomials over GF(3)

A monic cubic over GF(3) is irreducible iff it has no root in $\{0, 1, 2\}$. There are eight such polynomials; their companion matrices have order 26 or 13 in GL$_3$(GF(3)) **[B]**:

Table 1: Irreducible cubics over GF(3) and their order

Label	Polynomial	Order
f_1	$x^3 + 2x + 1$	26
f_2	$x^3 + 2x^2 + 1$	26
f_3	$x^3 + 2x^2 + x + 1$	26
f_4	$x^3 + x^2 + 2x + 1$	26
g_1	$x^3 + 2x + 2$	13
g_2	$x^3 + x^2 + 2$	13
g_3	$x^3 + x^2 + x + 2$	13
g_4	$x^3 + 2x^2 + 2x + 2$	13

The number four is forced: GF$(27)^*$ has $\varphi(26) = 12$ primitive elements in Frobenius orbits of length 3, hence $12/3 = 4$ minimal polynomials. Doc. 341 fixes only order 26 for the leptons, not a class; the assignment of charged leptons to the Dirac layer $\{f_2, f_3\}$ and neutrinos to the Majorana layer $\{f_1, f_4\}$ follows in Doc. 343, Theorem D‴ **[S]**.

The five classes

In G$(6) \cong M_8$(GF(9)), χ splits over GF(3) into factors whose degrees sum to 8. With cubic and linear factors this gives five classes **[B]**:

Table 2: χ factorisation classes with GF(27) content in G(6)

Type	$\chi(x)$	Count	Order
Doc.-341 type	$f_i^2\,(x+1)^2$	4	26
Mixed primitive	$f_i\,f_j\,(x+1)^2$, $i \neq j$	6	26
Order-13 type	$g_i^2\,(x+1)^2$	4	13 (sign partners of the f_i, Doc. 343)
Primitive×13	$f_i\,g_j\,(x+1)^2$	16	26
Thin	$f_i\,(x+1)^5$	4	26

The four primitive polynomials are the four Frobenius conjugacy classes $\{g, g^3, g^9\}$, $\{g^5, g^{15}, g^{45}\}$, $\{g^7, g^{21}, g^{63}\}$, $\{g^{11}, g^{33}, g^{99}\}$ of a generator g of GF$(27)^*$. Assigning the leptons to f_1 in Doc. 341 is a choice within this class; whether f_2, f_3, f_4 carry further mass layers remains open **[S]**.

.3 Theorem B: Entry rule of the harmonic primes

Doc. 057 describes primes as elementary relational steps: 2 octave, 3 fifth, 5 major third, 7 natural seventh, 11 eleventh, 13 thirteenth. A p-limit system admits primes up to p as interval building blocks.

Table 3: Order of GF$(3^k)^*$ and entry of new primes

k	|GF$(3^k)^*$|	prime factors	new	harmonics
1	2	2	2	octave
2	8	2^3	—	GF(9), Matzke's $\mathbb{Z}_3\mathbb{C}$
3	26	$2 \cdot 13$	13	thirteenth; leptons (Doc. 338)
4	80	$2^4 \cdot 5$	5	major third; Doug's G(3) orders
5	242	$2 \cdot 11^2$	11	eleventh; neutrinos (Doc. 340)
6	728	$2^3 \cdot 7 \cdot 13$	7	seventh; with 13 on one carrier
7	2186	$2 \cdot 1093$	1093	no small harmonics left
8	6560	$2^5 \cdot 5 \cdot 41$	41	—
12	531440	$2^4 \cdot 5 \cdot 7 \cdot 13 \cdot 73$	73	all four united

Entry rule [B]: A prime $p \neq 3$ divides $3^k - 1$ iff ord$_p(3) \mid k$; it therefore first enters at $k =$ ord$_p(3)$. This yields the order $13, 5, 11, 7$ for $k = 3, 4, 5, 6$. All four harmonic primes above 3 enter between $k = 3$ and $k = 6$; from $k = 7$ on, only primes ≥ 41 appear (with 1093, a Wieferich prime, at $k = 7$).

This is the algebraic reason why practically relevant harmonics end at the 13-limit, and why the FFGFT corpus has so far needed only the primes $\{2, 3, 5, 7, 11, 13\}$. The neutrino eleven of Doc. 340 ($\Delta m^2_{\text{atm}} = (11^2 - 1)\, m_\nu^2$) sits at $k = 5$; this had not been located in the corpus before.

$\mathbb{Z}_3$ compatibility: for $p \equiv 1 \pmod 3$, $\mathbb{Z}_3 \subset$ GF$(p)^*$. This holds for 7 and 13, not for $2, 5, 11$. The two $\mathbb{Z}_3$-compatible primes are exactly those that appear as structural numbers in Doc. 289 (top quark) and Docs. 338/341 (leptons) **[B]**.

.4 Theorem C: Chords as partitions of 8

For semisimple $X \in M_8(\mathrm{GF}(9))$ with χ split over GF(3) into factors of degrees $d_1 + \cdots + d_r = 8$, the eigenvalues lie in GF(3^{d_i}) and ord(X) divides lcm$(3^{d_i} - 1)$ **[B]**. A partition of 8 is thus the algebraic form of a chord of overtones $k \in \{d_i\}$; the order is the common period.

Table 4: Partitions of 8 and maximal semisimple order

Partition	lcm$(3^{d_i}-1)$	reading
$(3, 3, 1, 1)$	26	pure GF(27) chord (Doc. 341)
$(3, 3, 2)$	104	$= 2^3 \cdot 13$: GF(9) with GF(27)
$(4, 3, 1)$	1040	$= 2^4 \cdot 5 \cdot 13$: third with thirteenth
$(6, 2)$, $(6, 1, 1)$	728	$= 2^3 \cdot 7 \cdot 13$: seventh with thirteenth
$(5, 3)$	3146	$= 2 \cdot 11^2 \cdot 13$: eleventh with thirteenth
(8)	6560	$= 2^5 \cdot 5 \cdot 41$: maximal

The number of irreducible resp. primitive polynomials of degree k over GF(3) grows quickly (Möbius formula): degree 3: $8/4$, degree 4: $18/8$, degree 6: $116/48$, degree 8: $810/320$. The space of possible chords is therefore large — which leads directly to the question of discriminating power.

.5 Theorem D: Discriminating power and the boundary of the Galois layer

Density of Galois products

Forming all products of up to three building blocks $\{3^k, 3^k - 1, \text{prime divisors}\}$ for $k \le k_{\text{max}}$ with exponents $\{-1, 1, 2\}$, the mean logarithmic spacing of neighbouring values in the window $10 < x < 10^4$ drops as follows **[B]**:

Table 5: Mean log spacing of Galois products

k_{max}	values in window	mean spacing	expected chance hits at $0.5\,\%$
2	54	$12.8\,\%$	0.08
3	218	$3.17\,\%$	0.32
4	650	$1.06\,\%$	0.94
5	1423	$0.49\,\%$	2.06
6	2503	$0.28\,\%$	3.62
8	5311	$0.13\,\%$	7.69

From $k_{\text{max}} \approx 5$ on, the product set is denser than $0.5\,\%$: every number is "hit". A control test confirms this: at $k_{\text{max}} = 6$, ten physical mass ratios have 2.9 hits on average, two hundred random numbers 3.7. The hit rate is at chance level **[B]**.

The threshold: Galois grid versus fractal correction

Doc. 338 quantifies the fractal-recursive higher-winding correction between bare value and measurement at $1.05\,\%$ for $(m_\mu/m_e)^2$. The Galois grid is coarser than twice this correction up to $k = 3$ ($3.17\,\% > 2.10\,\%$) and hence discriminating; from $k = 4$ on it lies below ($1.06\,\% < 2.10\,\%$) **[B]**.

Consequence. The exact Galois layer ends structurally where the fractal correction begins. Up to GF(27) the algebra is exact and the $1\,\%$ is a higher-mode correction; above that, the algebra itself is finer than the correction and dissolves into the higher winding modes. Galois products with $k \geq 4$ are statistically indistinguishable from higher torus modes.

This is not a contradiction. It is the same mechanism as in the overtone series: the interval $(n{+}1) : n$ shrinks from 1200 cents ($n = 1$) via 105 cents ($n = 16$) to below 5 cents at $n \approx 250$; the series becomes quasi-continuous. Doc. 159 describes exactly this mode convergence on the fractal torus; here it appears in algebraic form.

Methodological rule

- $k \leq 3$ (GF(3), GF(9), GF(27)): identity layer. A hit with few building blocks is significant and may carry **[K]**/**[B]** if grounded geometrically in ξ (as 43200 in Doc. 338).
- $k = 4$ (GF(81), 5-limit): transition region; only with additional justification.
- $k \geq 5$: no status from product search. A relation holds here only if derived geometrically from ξ.

.6 Negative result: tau sector

At a tolerance of $0.05\,\%$, none of the ratios m_τ/m_μ, m_τ/m_e or their squares is a product of up to three Galois building blocks with $k \leq 6$ **[S]**. Hits at $0.5\,\%$ exist (e.g. $7 \cdot 11^2/3$ for $(m_\tau/m_\mu)^2$ at $0.17\,\%$) but lie at chance level by Theorem D. The tau ratio remains grounded in the corpus through winding numbers (Doc. 258, Doc. 310), not through a Galois identity.

.7 Summary

Table 6: Results and status

Result	Content	Status
A	8 irreducible cubics, 4 primitive; 5 χ classes	**[B]**
B	entry rule $k = \mathrm{ord}_p(3)$; order $13, 5, 11, 7$	**[B]**
B′	$7, 13$ are $\mathbb{Z}_3$-compatible ($p \equiv 1 \bmod 3$)	**[B]**
C	chords = partitions of 8, order = lcm	**[B]**
D	Galois grid $< 2\times$ fractal correction from $k = 4$	**[B]**
D′	hit rate physical = random for $k \geq 5$	**[B]**
—	tau sector no Galois product at $0.05\,\%$	**[S]**
—	physics of the classes f_2, f_3, f_4	**[S]**

Check scripts

Directory `2/python/Dok342_Skripte/`, wrapper `run_all_342.sh`:

- `pruef_342_faktorisierung_primzahlen.py` — Theorem A, p-limit assignment
- `pruef_342b_hoehere_harmonien.py` — Theorems B, C; product search with look-elsewhere note; Koide

- `pruef_342c_dichte_hoehere_harmonien.py` — Theorem D; density, random control, overtone analogue, threshold

All assertions passed.

Note on the use of AI assistance

This document was prepared with the assistance of Claude (Anthropic). All counts and order determinations are verified by Python check scripts with exact arithmetic. The physical interpretations and conclusions are those of Johann Pascher (ORCID 0009-0000-6518-4064).

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